Project Management Protocol of the Netherlands eScience Center

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Date	Version	Changes
1 July 2026	4.0	Major changes, renew protocol after 2025 restructuring.
1 June 2025	3.5	redacted internal references, added RSD checklist appendix and missing intro text, included Zenodo references, fixed tables, and improved description of annual review meeting
1 December 2024	3.0	Planning, Legal agreements, Project budget breakdown, change in the workshops and end report processes, open-access publications
1 October 2023	2.0	DT and PM mandates, Roles of Directors, PM role in Ambition 2, SS, KD and external projects, Communications role, Editorial team, breakdown of hours, opportunities beyond project, Technology Status Report, End report
23 September 2022	1.0	initial version

Contents

1 Scope and definitions

1.1 Scope

This document is the official project management protocol of the Netherlands eScience Center. It describes the activities, responsibilities, and procedures required to manage an eScience project after it has been formally awarded, through to its formal closure.

The protocol primarily applies to projects awarded through the eScience Center calls for proposals, although it also provides guidance for other types of projects as defined in Section 1.2. It follows the project life cycle and, for each phase, specifies the required activities, optional activities, responsible roles, and expected outcomes.

The call process itself is governed by a separate protocol [1] and is therefore outside the scope of this document. This protocol starts once a project has been formally awarded and covers all activities until the project is formally closed.

The following topics are also outside the scope of this protocol:

- independent project evaluation (impact, outputs, process, and collaboration);
- detailed guidance on project management tools (e.g. Ganttic, Metabase dashboards, AFAS);
- general project management methodologies and techniques.

The Management Team (MT) approves this protocol and reviews it annually. The document is organised according to the project life cycle (Section 1.4), presenting activities in chronological order.

1.2 Types of projects

The eScience Center receives an annual budget from NWO and SURF , the larger part of which is allocated to projects submitted by researchers working at eligible research performing organizations in the Netherlands in the form of the in-kind provision of Research Software Engineers (RSEs). The eScience Center also participates in externally funded projects and carries out internally funded strategic initiatives.

These projects differ in their funding model and governance. Throughout this protocol, they are grouped into the categories shown in Table 1.

Table 1: Project categories covered by this protocol

Project type	Description	Governed by
Call projects	Projects funded through the Netherlands eScience Center calls for proposals and delivered through in-kind RSE support.	This protocol
External projects	Projects funded by external organisations (e.g., NWO, Horizon Europe, industry). External contractual obligations take precedence where applicable.	This protocol + external agreements. See Appendix A for further details.
Other projects	Internal initiatives such as Research and Development (R&D) projects, Fellowship projects, and Dissemination and Community (D&C) initiatives.	This protocol where applicable + dedicated internal procedures. See Appendix B for further details.

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1.3 Stakeholders

An eScience project is a project involving the eScience Center, where responsibility is shared between different stakeholders who each have their own roles and responsibilities during specific phases of the project.

Table 2: Project stakeholders and their roles

Stakeholder	Role in the project	Further information
Lead Applicant (LA)	Represents the applicant organisation and is the primary external contact for the project. Accountable for the scientific contribution and the participation of the applicant project team.	Responsibilities may also be defined by the call conditions, grant agreement, or consortium agreement.
Lead Research Software Engineer (Lead RSE)	Coordinates the eScience Center's project activities, supports the successful delivery of the agreed work, and serves as the primary contact for the project.	See the formal **Lead RSE** role description in Appendix C.
Research Software Engineer, assigned to a project (RSE)	Contributes research software engineering expertise and carries out agreed project activities under the coordination of the **Lead RSE**.	See the RSE role and job descriptions [2, 3].
Consulting Research Software Engineer (Consulting RSE)	Provides specific expertise to support concrete technical aspects of a project.	Works under the coordination of the **Lead RSE** and is involved when additional expertise is required.
Section Head (SH)	Provides line management for the eScience Center project team, ensures appropriate staffing and capacity, and is accountable for the successful delivery of the eScience Center's contribution to the project.	Assigns the **Lead RSE** and project RSEs, and oversees team capacity and personnel matters. See Section 2.3.
Programme Officer (PO)	Coordinates the call, review, and project selection process, and supports the administrative management of the programme throughout the project lifecycle.	The funding decision communicated by the Programme Officer marks the starting point of this protocol.
Secretary	Provides administrative support for project coordination and formal meetings.	Assists with meeting organisation, documentation, and project communications.
Policy & Compliance Officer (P&CO)	Provides guidance on legal, policy, and compliance matters related to the project, including agreements with external partners.	Supports the preparation and review of collaboration agreements, NDAs, licences, and other legal documents where required.
Community Manager (CM)	Supports community building, engagement, and outreach activities related to the project.	Provides advice on developing community engagement plans and connecting projects with relevant research communities.

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Table 2: Project stakeholders and their roles (Continued)

Stakeholder	Role in the project	Further information
Communications	Supports the communication and dissemination of project activities and outcomes.	Advises on communication strategies and promotes project achievements through appropriate communication channels.
Performance Manager (PM)	Oversees portfolio performance, capacity planning, and portfolio reporting using project information and financial data. Supports MT decision-making on resource allocation and annual budgeting.	Operates at portfolio level rather than individual project level.
Management Team (MT)	Provides strategic oversight and approves project decisions that fall outside the delegated authority of the **SH**.	Includes the General Director, Section Heads, Portfolio Manager, and Performance Manager. The General Director is the final decision-maker.
eScience Center project team	The team responsible for delivering the eScience Center's contribution to the project.	Comprises the **Lead RSE**, RSEs, Section Head, and other contributors involved on behalf of the eScience Center.
Project team	The complete team responsible for delivering the project.	Comprises the eScience Center project team together with the Lead Applicant and the collaborating project partners.

1.4 Project lifecycle overview

Figure 1: Project Lifecycle Overview

At the eScience Center, a standard project lifecycle consists of three phases (see Figure 1) initiated, followed by the execution phase, during which research software is developed and project activities are coordinated and monitored. Finally, the project is formally closed through a structured closing process.

The following sections describe each phase of the project lifecycle in detail.

2 Project Initiation

The project initiation phase begins once a project has been approved for funding. Following the communication of the awarded project by the Program Officer (PO), the responsible **SH** initiates the project setup. The purpose of this phase is to establish the project within the eScience Center by completing the required administrative setup, assigning responsibilities, and preparing the project for execution.

The exact initiation process depends on the project type. External projects, as well as certain internal projects, may require additional contractual or legal arrangements before they can commence. Further information on the different project types is provided in Appendix B and Appendix A.

Once the project has been registered, staffed, and all applicable administrative, financial, and legal requirements have been fulfilled, it is ready to enter the execution phase.

2.1 SH assignment

Each project has one assigned **SH**, who is accountable for the project throughout its lifecycle.

For projects funded through the eScience Center's regular funding calls, the assigned **SH** normally corresponds to the section under which the proposal was submitted. In other cases, or where the assignment is not self-evident, the Section Heads jointly determine the most appropriate assignment, taking into account the project's scope, expertise, and organizational responsibilities.

Once the assignment has been agreed, the responsible **SH** is recorded and Finance is notified to initiate the project registration process.

Responsible: Section Heads.

2.2 Project registration

The table below summarizes the main project registration outputs, the responsible stakeholder, and the associated activities.

Table 3: Project registration activities and responsibilities

Output	Responsible	Main actions
Project record	Finance	• Creating the project code in AFAS • Assigning the responsible **SH** as budget holder
Project portfolio	Finance	• Creating the project portfolio folder [4] following the rquired template sub-folders • Requesting missing project documentation • Uploading the grant package (proposal, awarding letter, signed start form, and supporting documentation submitted by the LA)

Responsible: Finance.

2.3 Project staffing

The responsible **SH** is responsible for staffing the project, including the appointment of a **Lead RSE** and the assembly of the eScience Center project team [5]. Staffing decisions take into account the project's requirements, available capacity, and the expertise required for successful delivery. Staffing may be adjusted throughout the project as needed (see Section 3.3.3). The **Lead RSE** serves as the primary point of contact for the LA, unless otherwise agreed.

2.3.1 Staffing preparation

The responsible **SH** reviews the project proposal to determine the staffing requirements for the project. The assessment considers the project scope and objectives, the awarded budget, the expected timeline, the required scientific and technical expertise (making use of available organizational information on technology capabilities, e.g., technology skills surveys), any applicable contractual, legal, or organizational constraints, and the available capacity within the section based on the organizational planning system. These considerations form the basis for assigning the Science Center project team.

2.3.2 Team assignament

Every project must have: one accountable **Lead RSE**; and at least one additional Research Software Engineer assigned to the project. The eScience Center project team is assigned according to the following process:

1. The responsible **SH** identifies a suitable candidate for the **Lead RSE** role, taking into account the staffing preparation described above.
2. The **SH** discusses the proposed assignment with the candidate.
3. The proposed staffing is reviewed and confirmed by the Section Heads.
4. The responsible **SH** formally assigns the **Lead RSE** and the remaining project team members.

Once the **Lead RSE** has been assigned, they create the project page in the Research Software Directory (RSD, see also Appendix D). The RSD page serves as the central location for project information, including project description, team members, software outputs, publications, and other research products. The page should be kept up to date throughout the project lifecycle by the **Lead RSE**.

2.3.3 Project planning

Following project registration, the project is synchronized to the organization's planning system. The responsible **SH** records the **Lead RSE** in the corresponding project in Ganttic. The **Lead RSE** prepares the initial project planning by allocating the assigned team members across the different project phases. The planning serves as the baseline for project execution and may be updated during the project as required.

Responsible: SH.

2.4 Administrative kick-off meeting

Once the project has been registered and staffed, the responsible **SH** organizes an administrative kick-off meeting with the LA. Prior to the meeting, the **SH** and **Lead RSE** review the project proposal and identify any project-specific requirements that should be discussed with the LA. The purpose of the meeting is to introduce the collaboration with the Netherlands eScience Center and to ensure that the project is administratively and organizationally ready to start. These may include:

- any additional organizational or technical support required for the project;
- deviations from the default intellectual property or software licensing arrangements;
- data readiness, data access requirements, and any personal data or GDPR requirements;
- software sustainability expectations; and
- ethical, legal, or other project-specific risks.

The meeting is supported by a standardized administrative kick-off presentation available in the project portfolio under the *Templates projects* folder [4]. The presentation is periodically updated to reflect the requirements of different funding calls and the eScience Center's current procedures and policies. The responsible **SH** uses this presentation to ensure that all standard topics are covered consistently across projects while adapting the discussion to the specific needs of the project.

The presentation typically covers the following topics:

- Introduction to the Netherlands eScience Center (mission, organization, and expertise);
- Working with the eScience Center (roles and responsibilities, collaboration model, in-kind and in-cash contributions);
- Project lifecycle and project governance;
- Community activities and research impact (Research Software Directory, Digital Skills Programme, Fellowship Programme, etc.);
- Intellectual property, software licensing, publication practices, and open science principles;
- Project-specific needs, including scientific and technical expertise, SURF infrastructure requirements, and additional resources (e.g., AI subscriptions or other project-specific services).

The presentation used during the meeting and any meeting notes are stored in the corresponding project portfolio folder together with the other project documentation. The Secretary may assist the **SH** with scheduling the meeting.

Table 4: Administrative kick-off meeting overview

Timing:	Soon after project registration and staffing have been completed.
Participants:	**SH** (organizer), LA, **Lead RSE** (optional). Others may be invited as needed.
Objective:	Introduce the collaboration with the Netherlands eScience Center, discuss the project setup, and identify any project-specific administrative or organizational requirements before project execution begins.
Duration:	1 hour.
Location:	Online.

Responsible: SH.

2.5 Kick-off meeting

Following the administrative kick-off meeting, the **Lead RSE** organizes the project kick-off meeting with the project team, comprising the eScience project team, the LA, and the LA's team. The purpose of the meeting is to establish a shared understanding of the project objectives, work plan, responsibilities, and expected deliverables before project execution begins.

Prior to the meeting, and based on their review of the project proposal, the **Lead RSE** and the eScience Center project team identify any scientific or technical aspects that should be discussed with the LA. These may include:

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- the scientific objectives and expected project outcomes;
- the proposed work packages, milestones, timeline, and overall feasibility;
- technical challenges, assumptions, implementation risks, and whether additional technical expertise or consultation from other RSEs is required;
- software sustainability considerations and the initial Software Management Plan (SMP);
- dependencies on data, infrastructure, or external collaborators; and
- any issues requiring clarification before project execution starts.

The meeting is supported by a standardized project kick-off agenda available in the project portfolio under the *Templates projects* folder [4]. The **Lead RSE** uses this agenda to ensure that all standard topics are covered consistently across projects, while adapting the discussion to the specific needs of the project.

The agenda typically covers the following topics:

- Round of introductions;
- Project introduction and scientific objectives;
- Review of the project work plan;
- Agreements on the initial project planning and deliverables;
- Software sustainability and the Software Management Plan (SMP);
- Any other business.

The meeting outcomes, agreed action items, and any presentation material are stored in the corresponding project portfolio folder. The **Lead RSE** updates the project planning accordingly and prepares the project for the execution phase. These documents establish the initial project baseline and provide the starting point for the project log maintained during the execution phase. The Secretary may assist the **Lead RSE** with scheduling the meeting.

Following the kick-off meeting, the **SH** and **Lead RSE** determine, together with the Communications team where appropriate, whether internal or external communication activities should be initiated. These may include announcing the project on the eScience Center website, preparing news items for the kick-off meeting, or coordinating other outreach activities.

Table 5: Project kick-off meeting overview

Timing:	Shortly after the administrative kick-off meeting.
Participants:	**Lead RSE** (organizer), **SH**, LA, project team. Others may be invited as needed.
Objective:	Establish a shared understanding of the scientific objectives, work plan, responsibilities, and expected deliverables before project execution begins.
Output:	Action items agreed planning, initial project baseline.
Duration:	1.5 hours.
Location:	At the eScience Center or the project location.

Note: The **Lead RSE**, responsible **SH**, and LA are expected to attend the meeting in person at the eScience Center or the project location. Other project team members may participate online if appropriate.

Responsible: Lead RSE.

2.6 Legal agreements

The eScience Center promotes open, reproducible science and open-source software development. Projects funded through the eScience Center are carried out under the eScience Center Terms and Conditions [6]. However, project partners may require additional legal agreements to grant access to their facilities, data, infrastructure, or other resources. These agreements may also contain provisions related to intellectual property, confidentiality, data protection, or information security.

RSEs must not sign any formal agreements without consulting the responsible **SH**. Such documents include, but are not limited to:

- Guest agreement
- Data sharing agreement
- Non-disclosure agreement
- Consortium agreement
- Collaboration agreement
- Intellectual property (IP) agreement

The **SH** reviews the proposed agreement, consulting the **Lead RSE** where project-specific input is required, and determines whether additional review is required. If needed, the **SH** consults the Policy & Compliance Officer (P&CO) before the agreement is signed.The final signed agreement is archived in the corresponding project in the Project portfolio (see [4]).

Responsible: SH.

2.7 Technology plan

Every eScience project involves technological choices that influence software development, data management, reproducibility, and long-term sustainability. Following the project kick-off meeting, the **Lead RSE** prepares a technology plan in collaboration with the Lead Applicant and the project team. The technology plan is stored in the corresponding project portfolio (see [4]) and provides a shared reference for project execution.

The technology plan typically describes:

- selected technologies and rationale;
- software architecture or implementation approach;
- expected software features and data outputs;
- quality standards;
- reproducibility and sustainability considerations;
- key technical assumptions or constraints.

The technology plan is maintained as a living document and may be updated as the project evolves to reflect significant technological decisions agreed between the **Lead RSE** and the Lead Applicant and project team. An example technology plan is available in [7].

The **Lead RSE** is encouraged to consult colleagues with relevant technical expertise when preparing the technology plan. Where appropriate, the Community Manager (CM) may advise on software dissemination, community engagement, and strategies to promote software reuse and adoption. Likewise, the eScience Center's Special Interest Groups (SIGs) may provide guidance on domain-specific technologies, software engineering practices, and other technical topics relevant to the project.

Table 6: Technology plan overview

Prepared by:	**Lead RSE**, in collaboration with the LA and the project team. Where appropriate, colleagues with relevant technical expertise, the Community Manager, and Special Interest Groups (SIGs) may provide advice.
Target audience:	Project team.
Timing:	Shortly after the project kick-off meeting and before substantial software development begins.
Objective:	Document the technological decisions agreed for the project, including the selected technologies, implementation approach, software quality considerations, and other technical decisions that guide project execution.
Format:	Free-format document, maintained as a living document throughout the project as needed.

Responsible: Lead RSE.

3 Project execution

Once the project has been initiated, the execution phase focuses on delivering the agreed research objectives while monitoring progress, resources, risks, and deliverables. This is typically the longest phase of an eScience project and involves close collaboration between the eScience Center project team and the LA's team.

The **Lead RSE** coordinates the day-to-day technical execution of the project and facilitates regular meetings with the project team and the LA. The **SH** oversees project delivery from the eScience Center's perspective and supports the project when administrative, staffing, financial, or organizational intervention is required. The **Lead RSE** monitors progress to ensure that technical work, milestones, and deliverables remain aligned with the project objectives.

Because research projects vary considerably in scope, size, and complexity, the eScience Center does not prescribe a single project management methodology ([8–13]). Project teams are encouraged to adopt the practices and communication approaches that best suit their collaboration.

3.1 Project development

Project development is a collaborative effort between the eScience Center project team and the LA's team to achieve the scientific objectives defined during project initiation. Throughout the execution phase, the **Lead RSE** coordinates the technical work, facilitates communication between all team members, and ensures that development activities remain aligned with the agreed project objectives and planning.

Research software development is typically carried out iteratively. Scientific discoveries, user feedback, and technical challenges may require adjustments to the implementation approach. The **Lead RSE** and project team are expected to make informed technical decisions throughout the project, balancing scientific objectives with software quality, maintainability, sustainability, and the efficient use of available resources. Significant changes to the technical approach should be reflected in the Technology Plan (Section 2.7) and communicated to the project team.

Software developed at the eScience Center should follow the software development guide and established community best practices [8,14]. At the beginning of software development, the **Lead RSE** establishes the project public repositories and development workflow, ensuring that the project team has appropriate access and that the repositories are linked from the project's RSD page. Whenever appropriate, project repositories should be based on, and build upon existing resources available in the Netherlands eScience Center GitHub organization.

Responsible: Lead RSE.

3.1.1 Data, infrastructure and computing resources

Research projects may require access to computing infrastructure, data storage, cloud services, or specialized research facilities. The **Lead RSE** and the LA determine the project's infrastructure requirements during execution and ensure that appropriate resources are available throughout the project.

The Netherlands eScience Center primarily relies on the national research infrastructure provided by SURF, which includes high-performance computing (HPC) systems, cloud services, and data storage facilities. Depending on the project, access to SURF services may be obtained through project-specific allocations, the LA's institution, or dedicated test accounts. In addition, the eScience Center provides access to the DAS-6 distributed supercomputer.

Projects requiring new infrastructure or additional computing resources should consult the eScience Center's SURF liaisons, who can advise on the most appropriate services and the current application procedures. Up-to-date instructions and contact information are maintained on the eScience Center Intranet [15].

Responsible: Lead RSE (consults SURF liaisons where appropriate).

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3.2 Project meetings

Project meetings provide the primary forum for coordinating scientific and technical work, discussing progress, resolving issues, and planning future activities. Different types of meetings serve different purposes throughout the execution phase and should be adapted to the needs of the project.

3.2.1 Regular project team meetings

To keep the entire project team informed on project progress, the **Lead RSE** together with the LA organizes a periodical project meeting. The frequency and format depend on the complexity of the project and size of the project team.

Frequency:	Determined by the project team; commonly every two to four weeks.
Participants:	**Lead RSE** (organizer), LA, RSEs, and relevant members of the LA team.
Objective:	Coordinate work, assess progress, resolve issues, and update planning.
Output:	Agreed actions and relevant decisions recorded in the project log (see Section 3.3.1).
Duration:	Typically 30–60 minutes.
Location:	In person or online, as agreed by the project team.

The agenda of this meeting should include:

- Status update from all stakeholders
- Discussion of scientific progress
- Discussion of technological progress, issues and choices
- Alignment of project progress with the project workplan, and adjustment of the latter, if necessary.

Responsible: Lead RSE (co-organizer with the LA).

3.2.2 Internal coordination and escalation

The **Lead RSE** and the responsible **SH** maintain regular communication throughout the project. Internal coordination may take the form of scheduled meetings or ad hoc discussions, depending on the complexity and needs of the project. The **Lead RSE** keeps the responsible **SH** informed of the project's progress and of significant issues that may affect its successful delivery.

Internal coordination may cover topics such as:

- progress against the project workplan and agreed deliverables;
- significant changes to the Technology Plan (Section 2.7);
- project risks, issues, and mitigation actions;
- staffing, resource allocation, and budget monitoring;
- workshops, community engagement, and communication activities;
- opportunities for collaboration with other projects or reuse of software; and
- other matters requiring organizational support or a management decision.

Issues that cannot be resolved within the project team, or that materially affect the project's objectives, budget, timeline, staffing, contractual obligations, or collaboration, should be escalated to the responsible **SH**. Where appropriate, the **SH** may involve other stakeholders, such as Finance, Communications, CM, P&CO, or the MT, to support decision-making or resolve the issue.

Responsible: Lead RSE and SH.

3.2.3 Workshops and community events

Some projects include workshops or community events as part of their agreed project activities. Their purpose is to foster collaboration, engage research communities, disseminate project results, collect user feedback, or support the adoption and sustainability of research software. The organization, funding, and expected outcomes depend on the applicable funding programme and project agreement.

When a workshop is funded through the project budget, the following process is followed:

1. The LA prepares a scientific agenda and a budget proposal using the project templates available in the *Templates projects* folder [4]. The **Lead RSE** supports the preparation of the workshop and contributes to the scientific agenda.
2. The responsible **SH** reviews the proposed workshop and approves the scientific programme.
3. Finance reviews the proposed budget and confirms that the planned expenditure complies with the project conditions.
4. Finance notifies the LA of the budget approval and initiates the corresponding financial process.
5. After the workshop, the LA submits the final financial documentation, including the corresponding invoices, to Finance.
6. Finance verifies the final expenses and processes the remaining payment.

The CM may support the project team in designing workshop agendas and engaging the target community to maximize the long-term impact of the workshop outcomes.

eScience Center–Lorentz Center Competition workshops

Projects awarded through the joint eScience Center–Lorentz Center Competition differ from regular project workshops. The workshop proposal is developed and submitted as part of the competition, and the role of the eScience Center project team is defined in the awarded proposal.

The **Lead RSE** and the assigned RSEs support the scientific organizers throughout the preparation, delivery, and follow-up of the workshop in accordance with the agreed project plan. Typical contributions may include scientific and technical advice, workshop organization, development of research software or demonstrators, and the dissemination of workshop outcomes, such as publications, software releases, white papers, or follow-up funding proposals.

Responsible: Lead RSE and LA.

3.3 Project monitoring

Project monitoring provides the information needed to assess whether the project is progressing according to the agreed objectives and workplan. Throughout the execution phase, the **Lead RSE** monitors project progress, maintains the project log, and, together with the responsible **SH**, ensures that emerging issues are identified early and addressed in a timely manner.

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3.3.1 Project log

The project log is the chronological record of significant project events, decisions, and agreements throughout the execution phase. The **Lead RSE** is responsible for maintaining the project log, ensuring that it is shared with the LA, and storing it in the *B – General documents* folder of the corresponding project in the Project portfolio (see [4]).

The project log may be maintained in any suitable shared platform (e.g., a shared document, project wiki, or other collaborative environment), provided that the version stored in the Project portfolio remains up to date. Rather than duplicating information, the project log should provide references to the authoritative project documents, repositories, and other relevant resources. An example project log is provided in Appendix E.

Typical information recorded in the project log includes:

- key meetings, including dates, agreed actions, and important decisions;
- significant changes to project staffing, infrastructure or computing resources;
- workshops, conferences, and other relevant project events;
- links to the RSD project page, repositories, project deliverables, and other supporting documentation;
- updates to the Technology Plan and other project plans; and
- formal project reviews and agreed follow-up actions.

The project log should remain concise and avoid duplicating information already maintained elsewhere. Detailed technical descriptions, planning documents, and review reports should be maintained in their respective project documents, while the project log records when these changes occurred and where the corresponding information can be found.

Responsible: Lead RSE.

3.3.2 Progress, hours and budget monitoring

The **Lead RSE** is responsible for monitoring the overall health of the project throughout its execution. This includes tracking progress against the agreed workplan, monitoring the use of project hours, budget, and other project-specific resources, identifying emerging risks, and informing the **SH** and LA when adjustments are required. Such discussions are normally conducted through the internal coordination and escalation process (Section 3.2.2). To support these activities, the **Lead RSE** uses the eScience Center project management systems, including Ganttic for project planning, Metabase dashboards for project monitoring, and the financial information maintained by Finance.

The **Lead RSE** routinely monitors:

- progress towards milestones and agreed deliverables;
- project risks, dependencies, and issues affecting delivery.
- expenditure of project hours against the remaining project allocation;
- staffing and resource availability; and
- assess requests for the use of project-specific in-cash resources available for project expenses, such as workshops or open-access publication costs, where applicable;

Project information collected during execution also contributes to the periodic portfolio reviews carried out by the Section Heads and Performance Manager (PM) to support resource planning and annual allocations.

Responsible: Lead RSE, SH and Finance

Stakeholder	Responsibilities	Supporting systems
Lead RSE	• Monitor project progress and deliverables • Monitor project hours and resource allocation • Monitor project-specific cash resources • Identify and communicate risks and deviations • Escalate issues to the **SH** when required	Ganttic, Metabase
SH	• Oversee project health • Support corrective actions • Decide on organizational escalations	Ganttic, Metabase
Finance	• Maintain financial information • Confirm the available in-cash balance and eligibility of proposed expenses • Process approved expenses and payments • Provide updated budget information to the **Lead RSE** and **SH**	AFAS, Metabase

3.3.3 Risks, issues and project changes

During the project life cycle, the workplan may change substantially:

- New deliverables because of additional funding
- New workplan because of changes in the research goal and/or in the technology used
- Timeline, leading to a different end date.

Any of these changes needs explicit approval from the PM team or the DT.

Type of request	Decided by:
Budget requests within the PM mandate [15]	PM team
All requests regarding budget changes outside the PM mandate	DT (via PD)
Early termination	DT (and DT informs the Board)

For **external projects**, changes must follow the formal agreements (e.g., grant or consortium agreement). The PM handles extensions within their mandate or escalates to the DT. The PM informs the external funder or consortium of the outcome.

- Proposal changes request The LA must submit a formal request to the PM team (by email via the PM, in PDF format, signed) containing:
  - project title and project number
  - requested change (e.g., time/dates, RSE hours, scientific goal) and motivation for this change
  - conditions such as deliverables: any new deliverables should have updated timeline. If none, state explicitly.
  - any motivated budget change, such as
    * LA wants to increase their involvement
    * change in research personnel (if applicable in the case of older projects)
    * transfer between hardware and PYR for RSE or LA personnel costs (if applicable, for older projects)

    * any in-kind to cash change, or vice versa (including requests with the extra cash budget from the LA).
  - any prior or planned inactivity on the project, such as
    * personnel shortage on the LA side due to maternity/sick leave or hiring delays (but there is a concise timeline on the hiring procedure)
    * unavailability of RSEs
    * additional data that needs to be collected.
  - any delay with the start date.

- Processing the changes request and decision Upon receipt of the request, the PM assesses if the request should be granted based on considerations such as

  - whether the new objective is scientifically promising or technologically interesting? (if applicable)
  - collaboration status with the LA;
  - prior problems regarding the project;
  - benefits for the eScience Center (e.g., smooth wrap-up, future funding opportunity); and/or
  - availability of RSEs with relevant expertise to work on it.

  The PM can consult with RSEs and the TL on whether the new planning is feasible, and with Finance for a budget remaining after necessary recalculation. In case of additional funding, the DT (via the PD) will decide, after a budget calculation by the Finance and approval by the DoO. Otherwise, the PM puts the request on the agenda for the next PM meeting, containing:

  - the motivated request (uploaded to the project portfolio
  - the recommended action
  - the prepared decision on the PM meeting agenda.

  The PM team may request more information from the LA via the PM (and thus postpone the decision on the request). The LA can provide the new information via an additional PDF signed letter or as amendment to the original letter.

  After the final decision, the PM notes the official decision in the decision document. If the request is not approved, the PM communicates this to the LA. If the request is approved, the PM

  - coordinates with Finance to finalize the extension (budget/Exact update, extension letter, LA communication);
  - communicates the extension to the **Lead RSE**.

  The **Lead RSE** then

  - communicates the extension to the project team;
  - updates planning and adjusts staffing, if necessary;
  - ensures website and RSD are updated (e.g., if dates or affiliation changed); and

  If the request involves a DT decision, the PM submits a request formally through the PD.

- Conflict resolution and complaint procedure The eScience Center adheres to the Code of Conduct outlined in the Personnel Policy (cf. [16]). Conflicts involving eScience and the LA team on the projects should be addressed using this four-step process:

  1. The **Lead RSE** first seeks resolution with the LA; the PM may be consulted or join in discussion if needed.
  2. If unresolved, the issue is escalated to the PM (via the **Lead RSE**), who organizes a meeting to mediate.
  3. Persistent issues can be escalated to the PD via a written summary submitted through the PM.
  4. The PD may escalate to the DT if necessary.

If the issue concerns the PM, RSEs may escalate to their manager (**SH**). An external confidential advisor ('Vertrouwenspersoon') is also available for anonymous support [15]. Conflict resolution may lead to project adjustments, such as staffing changes (cf. Section **??**).

- Early project termination Early project termination may occur by mutual agreement, or be initiated by the LA or the eScience Center. In the first two cases, the LA submits a signed letter (via the PM) to the DT, requesting project termination, explaining the situation and proposing how to handle remaining project resources (e.g., RSE hours, cash contribution, FTE commitment of the LA, workshops, sustainability budget).

  The PM may request project termination if the LA or project partners violate the terms of Awarding Letter or Special conditions ("Bijzondere Voorwaarden"). The PM submits a letter with the explanation to the DT via the PD. If approved, the PM informs Finance, which finalizes the process (by making changes in Exact and preparing a termination letter).

Responsible: Lead RSE and SH

3.4 Project quality and outputs

3.4.1 Software quality and sustainability

Ensuring software quality and sustainability is integral to the code development process cycle at the eScience Center. All RSEs are expected to follow our software development guide and best practices [14] and the GenAI policy [17]. Code should be designed to be as generic and reusable as possible from the start.

As part of the annual review (see Section **??**), a code review is organized by the **Lead RSE**. Depending on the project it could take the form of a reusabilithon[1], a review of code on GitHub, or something else entirely (format to be approved by TL). The reviewers for this process are typically other RSEs at the eScience Center.

The goal is to review software of the project for:

- its usability (reproducing steps of installations, and running it on a machine/laptop)
- overall software quality and suitability
- whenever appropriate, correctness of code
- adherence to the technology plan (Section 2.7)
- adherence to eScience Center best practices
- opportunities for reuse of software in other projects
- correct inclusion in output systems (Section 3.4.4).

Reviewers make written suggestions for improvements, and flag major issues encountered. These issues serve as input for the TL for the formal annual review meeting (see Section **??**). These notes are stored in the project log.

Responsible: Lead RSE and project team.

[1] Coined by the Software Sustainability SIG, the term refers to a 2-3 hour session where a group of RSEs evaluates a piece of software for usability, provides feedback to the developers, and formulates recommendations to improve its (re)usability.

3.4.2 Data and Software Management Plans

For some projects, Data and Software Management Plans (DMP [18] and SMP [19]) outline how project data and software are maintained. Depending on the call, the LA must submit complete DMP and SMP documents within 6 months of the project. The **Lead RSE** may assist in drafting the DMP, which the LA submits to the PM, who asks a TL for review and approval. For call projects since 2021, SMPs are included in project proposals.

The LA is responsible for keeping both plans up to date and must inform the PM and TL of changes via the **Lead RSE**. The PM may request updates as needed, with support from the **Lead RSE**.

Responsible: LA and Lead RSE.

3.4.3 Project documentation

3.4.4 Project outputs

Project deliverables include research articles, presentations, talks, posters, tutorials, datasets, blog posts, white papers, and workshops, as well as software-related outputs such as code releases, software papers, demos, videos, tutorials, and training materials. RSEs strive to apply FAIR principles [20, 21] to all project deliverables. They should have

- (concept) DOIs from publishers or open-access archives like Zenodo, arXiv, or DANS;
- acknowledge the eScience Center project grant [2]; and
- listed RSEs working on the project as (co-)authors;
- adhere to the GenAI policy [17].

RSEs must record all project outputs in relevant systems and databases (see Table 7) to ensure knowledge transfer. The PM expects the team to follow the deliverables plan outlined in the proposal and workplan. RSEs contribute to publications and software/data releases.

Table 7: Output management

<table>
<tr><th>What</th><th>Responsibility of</th><th>Responsible for</th><th>URL</th><th>Additional info</th></tr>
<tr><td rowspan="2">Zenodo (NLeSC community)</td><td>PM / CMs</td><td>curating and approving new publications</td><td rowspan="2">Zenodo link</td><td rowspan="2">• Publications on Zenodo are added to the community</td></tr>
<tr><td>RSEs</td><td>getting a (concept) DOI for a data or software release, or a document (e.g., non-peer-reviewed publications)</td></tr>
<tr><td rowspan="2">RSD, software and project pages</td><td>PM / Lead RSE</td><td>• ensure the eScience team has maintainer access
• decide on reuse of the project page if the project continues under new funding.</td><td rowspan="2">See [22]</td><td rowspan="2">See Appendix D</td></tr>
<tr><td>RSEs</td><td>• create project page, if needed
• maintain pages with complete, up-to-date metadata</td></tr>
</table>

Continued on next page

[2] By adding the following sentence into the publication: "<full project title> is a project of the Netherlands eScience Center, funded under grant number <grantid>."

Table 7: Output management (Continued)

What	Responsibility of	Responsible for	URL	Additional info
Project portfolio	PM	adding a direct link to the project portfolio in Ganttic	cf. [4] for structure explanation.	An internal archive with regular automatic backups. Direct link to the portfolio folder available in Ganttic
	Lead RSE	ensuring that all docs are available		
	RSE	• upload outputs (e.g., papers, reports, presentations) • link the source material in the project log		
The eScience Center blog	RSEs	• Create a blog post about the project, its progress, or outcomes. • add final blog URL to project RSD page	Indexed at eScience Blog	Instructions on blogging are posted on the Intranet [15]
	Editorial Team	• Review blog post entry of project team • Help at any stage with writing and publishing process		

Open access publications and open source software are mandatory for all call projects. The PM and **Lead RSE** inform the LA if needed. Open access fees are budgeted by Finance annually in the call budget (with the PD approval); if no budget is available, alternatives can be explored:

- publishing through the LA's institution (either open access funds are available or the LA organization is connected to publish open access in a lot of journals through the library without a fee)
- choosing the best option for open-access publishing (is it a diamond or gold open-access journal)
- publishing closed access and self-archiving either a preprint or publication after six months under Taverne agreement in Dutch copyright law (cf. [23, 24]).

The **Lead RSE** and PM consult Finance regarding payments for an open access publication.

For **external projects** the expected deliverables are also part of the formal project documents (proposal, contract, etc. The **Lead RSE** is expected to keep the PM informed of the status of deliverables throughout the project.

Responsible: Project team.

3.5 Knowledge transfer and communication

To increase visibility of the project and its results, the project team (including RSEs, PM, TL), Communications, CMs, share knowledge and outcomes both inside and outside of the organization. The **Lead RSE** ensures that

- project results are timely shared with Communications, CMs, and relevant **SH**;
- project and software pages on the RSD are properly updated; and
- specific requests to facilitate project visibility are sent to Communications by RSEs.

Moreover, PMs, Lead RSEs and TLs work together to spot opportunities for cross project collaboration (e.g., by reusing software or knowledge in these projects or as a new reusability project, read more in Section 3.6.5).

TO BE MERGED: The **Lead RSE** promotes the project's visibility through project demonstrators, presentations, and other means. All RSEs are expected to communicate externally about the project and its deliverables, as outlined in

Section 3.4.4. Communications supports the project team by highlighting project through various channels, including but not limited to news items, social media posts, videos and interviews about relevant the scientific impact of the research software developed or used. The **Lead RSE** (or occasionally the PM) provides Communications with relevant information.

Blog posts are an optional but highly recommended output of the projects. Any team member, from LA to RSE, can author them. Topics may include simplified research summaries, tutorials on learned skills or technologies, or updates on workshops, publications, and other project outputs.

RSEs share project progress and results (e.g., technology plan, milestones, code releases) with colleagues, including SIG presentations. Each RSE prepares and updates a 3-slide deck or pitch [4]. The **Lead RSE** ensures a demonstrator is available after the first major software release.

The LA and their team are encouraged to participate in relevant Digital Skills Workshops [25] from the eScience Center. Furthermore, if the LA and their team require a project-specific training workshop, the **Lead RSE** involves

- workshop coordination with CMs, who can advise on organization and training material development; Finance handles any related payments if applicable;
- the PM discusses RSE hours spent on training organization and consults Finance if needed.

For **external projects**, consortium agreements (e.g., Non-Disclosure Agreement or NDA) may define result communication. The **Lead RSE** and PM review these at project start to clarify communication boundaries.

Responsible: Project team and Communications.

3.6 Project review

Project reviews follow an annual review cycle consisting of the following phases…

Scheduled:	Yearly
Stakeholders:	PM (organizer, chair), **Lead RSE**, LA, TL, optional: other project team members, **SH**
Purpose:	• to ensure that the project is still on track • to discuss any persistent issues to ensure optimal collaboration between project team • to explore opportunities beyond the project
Outcomes/Actions:	List of agreed actions and PM/TL advice for next steps.
Duration:	Max 1.5 hours
Location:	At the eScience Center or the project location[a]

[a] Mandatory participants must attend the meeting in person, while optional participants may join via video conference.

For call projects longer than one year, the PM organizes annual reviews. The details are described in the terms and conditions document [6]. For one-year projects, review meetings are optional and decided by the PM in consultation with the TL and **Lead RSE**.

Each review includes discussion and actions on reusability and sustainability of project results (as described in the DMP and SMP), status of collaboration, and follow-up opportunities. The agenda of this meeting is:

- introduction and purpose of meeting
- scientific goals status (the LA and team)
- project output status (the project team)

- collaboration status (including hours admin and bottlenecks)
- use of digital infrastructure and SURF support (if applicable)
- next steps and future opportunities.

For **external projects**, review meetings are typically part of the project process. The PM, with the **Lead RSE**, decides if a (lightweight) internal review is needed.

3.6.1 Review meeting preparation

	Stakeholder
Prepared by:	LA and **Lead RSE**, with optional input from other stakeholders.
Reviewed by:	PM accountable for project, TL accountable for technology.
Target audience:	PMs, TLs, **SH** and RSEs.

The **Lead RSE** coordinates preparations for the review meeting:

- request the PM to share the standard review meeting slide template [4] with the LA team
- requests project team members to contribute to the review meeting slides, wherever appropriate,
- compiles technology status report (see Section 3.6.2).

LA and **Lead RSE** jointly prepare the slides:

- 3-5 slides summarizing progress toward research goals. The LA is not expected to present published content or the original workplan or proposal. If applicable, the LA reports on workshops.
- 1-2 slides on the technology plan status, including reuse, adoption, and sustainability, supporting the technology status report.
- highlight any scientific or technical bottlenecks and assess if the project is on track or needs replanning;
- the PM reports on the RSE hour status (i.e. the number of hours already spent); and
- the project team provides a feedback on collaboration and suggest improvements if needed.

Additionally, the **Lead RSE**:

- together with the LA, ensures all output is properly registered (see Section 3.4.4), and adds missing URLs/DOIs to the slides;
- coordinates with the team to finalize the presentation at least 2 working days before the review meeting;
- uploads the slides to the project portfolio; and
- informs PM and TL that slides are ready and are in the project portfolio.

3.6.2 Technology status report

Prior to the annual review meeting, the PM asks **Lead RSE** to fill in the technology status report. This report summarizes the technical progress of the project and includes RSD project page with its deliverables, URLs to project plans, software quality and community involvement. The **Lead RSE** prepares it in collaboration with the project team; it serves as input for the TL during the review. The **Lead RSE**:

- collects output and relevant information from LA and team
- meets with CM (for the community outreach activities) to draft the technology status report
- draft the rest of the report with the TL

Two weeks prior to the review meeting, the **Lead RSE** submits the completed report to the TL. The TL conducts a code review or software health check based on the report and shares the results with the eScience team. If needed, the **Lead RSE**, TL, and PM may meet to discuss the findings before the annual review meeting.

The report and the review(s) are archived by the TL in the project portfolio.

3.6.3 At the review meeting

The time breakdown of the meeting agenda is follows:

- presentation by the LA (max. 20 min)
- presentation by the **Lead RSE** (max. 20 min), including the RSE roles and deliverables
- discussion (max. 40 min)
- summary, action points and conclusions.

The PM chairs the meeting and, with the TL, acts as reviewer. The TL highlights tech/software issues. Everyone contributes to the discussion. PM and TL assess deliverables, and comment on their status:

- have the objectives outlined in the proposal been sufficiently addressed? (PM)
- does the project follow the workplan in terms of deliverables? (PM)
- has the output been registered according to the rules of output management (Section 3.4.4)? (PM, TL)
- does all project output have publications (including software and data papers)? (PM)
- are there any issues flagged during the code review that needs to be discussed with the project team? (TL)
- does the project team sufficiently engage and align with relevant communities (e.g., via the workshops)? (PM)
- does the project adhere to the technology plan, SMP and DMP? (TL)

The eScience project team comments on any further possibilities for reusability, adoptability and sustainability of the software, and the project team comments on possible collaborations beyond the project.

The PM updates the slides with action points, agreements and plans (with the project partners agreement). The PM logs the meeting in the project log.

3.6.4 After the review meeting

PM shares the updated slide deck with the project team members to check the agreements written down. PM ensures that the final version of the presentation(s) uploaded to the project portfolio is correct.

3.6.5 Opportunities beyond the project

A project team can explore different opportunities for ideas that stem from the project that go beyond its scope and/or budget. Appendix **??** summarizes the role of PM in such projects. The PM discusses these opportunities with the project team during the review meeting.

- Increasing reusability (in this document called software sustainability) Some projects or calls have dedicated software sustainability budgets [26]. Until 2020, each project received its own Generalization ("Generalisatie") budget for reuse of project results. Since 2020, this budget is allocated at the programme or call level. The PM and **Lead RSE** must check the call text, awarding letter to confirm its availability.

  The PM signals potential for reuse to the TL, who discusses it with the **Lead RSE** and, if needed, the TL team. RSEs with ideas for sustainability projects can contact the TL or DoT. The process follows the SS protocol [15].
- Knowledge and Development For the development of broad and deep knowledge on digital technologies and their application, RSEs can apply for so-called Knowledge and Development (KD) project. The process is described by the KD protocol [15].
- External funding The project team may decide to pursue other funding opportunities. The eScience Center encourages RSEs to pursue external funding opportunities to promote the organization profile as research organization. PD oversees the external funding opportunities to promote the organization profile as research organization. PD oversees the Acquisition activities and the process. The relevant information is available via Intranet [15].
- Fellowship To stimulate community engagement lasting longer than project lifetime, the eScience Center funds annual Fellowship Program [27, 28]. The eScience project team is not eligible for this program, however, the LA and their team are.

Responsible: Project team and Communications.

4 Project closing

Project closing is the final phase of a project. In this phase the **Lead RSE** requests and previews the end report, the PM (with the help of Finance) processes the end report and accepts the project deliverables. Once the project is formally closed, RSEs can no longer write hours or work on this project.

4.1 Handling end report

All completed call projects at the eScience Center must have an end project report.

Written by:	the LA, assisted by the **Lead RSE**.
Target audience:	PMs, RSEs, Communications (layman summary), Finance (accountants), TLs
Schedule:	• written in last months of the project, • submitted 3 months after the project end at latest, • archived in the project portfolio.
Approved by:	The PM team and Finance.

For call projects, one month before the project end, the **Lead RSE** requests the LA to submit the scientific and financial end report ('Financieel en wetenschappelijk eindverslag'), providing the eScience Center template [4].

RSEs provide the LA with all necessary information for the end report.A complete end report includes:

- Public summary (in English) with objectives and results,
- Challenges (the team encountered during the project),
- Opportunities (the work of the project led to),
- Remarks on sustainability and latest management plans,
- Outcomes missing from the RSD (e.g., software, papers, presentations, workshops, testimonials),
- Information on the project workshops,
- Financial overview (if applicable),
- LA's signature and date (or financial/project manager's if LA unavailable).

For collaborative calls, the LA and **Lead RSE**'s report to other funders (e.g., NWO) suffices if it covers the above points.

Before the formal closing of the project, the **Lead RSE**:

- receives the signed end report from the LA (preferrably, in PDF format)
- verifies the scientific part of the end report against the end report checklist. If needed, the **Lead RSE** asks the LA for additional information or corrections to the end report, before submitting it to the PM.
- ensures all missing project output is registered in the appropriate systems (see Section 3.4.4)
- archives the end report, the lastest DMP and SMP, checklist in the respective project portfolio.

If the end report is satisfactory and all the aforementioned steps are completed, the **Lead RSE** submits it to the PM to get the project formally closed. The PM requests Finance to review the financial part of the report and either approve it or request corrections to it.

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Projects that are funded by software sustainability budgets have their own procedure for end reports (see Appendix **??**). The end report of these projects consists of output registered on the RSD project page and updated summary of the project.

For **external projects** the way a project is formally closed depends on the formal documentation for a project and requirements from the external funding organization. If an end report is required, the **Lead RSE** contributes to it (PMs can assist when needed), and care should be taken to reserve some time (and budget) during the project for this effort. Finance prepares the financial part of the report. The **Lead RSE** shares the final version of the end report written for the external party or funding organization (e.g., EU, NWO) with the PM and Finance, and archives it in the project portfolio. In any case, the **Lead RSE** ensures that the RSD project page is up-to-date (with complete deliverables and layman summary of the completed project).

Finance periodically sends a list of missing end reports to the PM team. If the end report is not submitted yet, the PM sends a reminder to the LA.

4.2 Formally closing the project

If both scientific and financial part of the end report is satisfactory, the PM puts decision to formally close project on the PM meeting agenda. After the formal decision[3], the PM notifies the TLs, Finance and Communications (with the links to the documents). Finance handles the approved reports and formalities related to closing the project. This includes getting the final signature by DoO or Executive Director on the official letter for the LA about the project closing ('Afsluitingsbrief'), communicating it to the LA, and archiving the letter in the project portfolio.

The PM ensures that the project is marked as complete on the

- RSD: The **Lead RSE** updates the project page with the lay summary (from the end report), any missing meta-data (e-Infra use, keywords, deliverables, etc.) and sets the project status to 'Closed'. See also Appendix D.
- Corporate page: Communications may also request more information to promote the completion of the project through various channels, including but not limited to a news item, social media post, video and interview.
- Exact/Project portfolio: Finance closes the Exact project budget, uploads the official closing letter to the LA, as well as all appropriated documents.
- Ganttic: The PM ensures that no one is assigned to the project in the future.

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Appendices

A Applying the protocol to external projects

Projects funded by external organisations (e.g., NWO, Horizon Europe, charities, industry, or public-private partnerships) are subject to contractual obligations defined by the funding body or collaboration agreement. Where these obligations differ from the guidance provided in this protocol, the external contractual requirements take precedence. This appendix describes how the Project Management Protocol should be applied to externally funded projects and highlights the main adaptations required to comply with external funding conditions while maintaining the eScience Center's internal project management practices.

1. External agreements take precedence * Grant Agreement * Consortium Agreement * Customer contract * Project-specific governance 2. What remains unchanged * Roles at the eScience Center * Internal reviews * Documentation * Quality assurance * Risk management * Project closure 3. Typical differences * Additional reporting * Deliverables * Milestones * Steering committees * Consortium meetings * External approvals 4. Examples * NWO * Horizon Europe * Industry projects

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B Applying the protocol to internal projects

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C Lead RSE role description

This role description includes guidelines that need to be followed by RSEs fulfilling the role; they will be complemented by protocols.

1. Role	**Lead RSE**
2. Place in the organization	Project
3. Contacts	**SH**, project team RSEs, external project partners
4. Purpose	To be responsible for the day-to-day running of a research project at the eScience Center and to act as the main point of contact for the project. Each project has one **Lead RSE**.
5. Main tasks & responsibilities	• Agreeing with the accountable **SH** on the division of tasks and the allocation of time within the project • Coordinating day-to-day activities with other RSEs working on the project • Making sure that agreed activities, procedures and targets are carried out and met on time • Monitoring project progress, including project hour expenditure, and regularly reporting progress to the accountable **SH** • Ensuring that major technological decisions are appropriately reviewed and monitoring their implementation, in consultation with the accountable **SH** • Ensuring that generalization and re-usability opportunities are considered and implemented from the start of the project, in consultation with the accountable **SH** • Ensuring the presence of the accountable **SH** project at meetings where their participation is required (e.g. project kick-off and review meetings) • Solving everyday technical and managerial problems and escalating them to the accountable **SH** when necessary • Making sure all project outputs are properly released, documented and archived in the designated systems • Ensuring the visibility of the project through project demonstrators, slide decks and other appropriate dissemination activities
6. Competencies	• Leading • Project management • Communicating • Cooperating • Result orientation • Decision making
7. Available resources (budget, hours, training)	In project budget
8. How to get this role	**SH** assigns **Lead RSE** based on skills, experience, knowledge, interest and availability

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D RSD project pages

Each project at the eScience Center has an RSD page.

Project Initiation At the project start, the following data should be added to the RSD project page:

- Grant ID: currently, the Awarding letter reference, for External funders (e.g., EU, NWO) often provide their own ID.
- Funder(s): list of all funding organizations, including the eScience Center.
- Dates: project start and end dates.
- Participating organizations: list of the organizations, including the eScience section(s).
- Research domain: relevant project domain, often from the proposal. Multiple domains can be chosen.
- Description: typically copied from the proposal abstract.
- A cover image: initially a placeholder image provided by the Communication.
- Team and maintainers: the **Lead RSE** and the project team members, including the LAs and their team, with roles and ORCIDs where possible. By default, **SH** is listed as a maintainer, while the inclusion of the LA is optional.
- Relevant eInfrastructures used.
- Relevant keywords and categories.
- eScience funding call name, if applicable.

During the project lifecycle, the **Lead RSE** is responsible for managing page updates.

Project Execution During the project, the following data should be added or updated as needed:

- repository link: URL to the GitHub (or other version control platform) containing the project's codebase.
- project team: the list of all project team members including the LAs and their team, with roles and ORCIDs where possible. RSEs should update the list as the team members change, and use the role field to indicate involvement periods.
- related projects: the link(s) to any precursor project(s) in RSD.
- deliverables: any project deliverables (see also Section 3.4.4) including but not limited to
  - all research deliverables of the project, i.e. publications such as papers or published datasets (using the DOI provided by the publisher), preprints of papers (using the DOI provided by the preprint server) self published research outputs, such as datasets, presentations, posters, reports, etc. (using the DOI it receives when uploaded to Zenodo), Online material such as websites, notebooks, online tutorials, blogs, videos, news items, interviews, etc. (registered using the URL and a short description).
  - all software deliverables: all reusable software should have its own RSD software page and be linked as related software. Single-use software (e.g., scripts) can also be archived or released via Zenodo and can be added via DOI. Versioned releases archived in Zenodo can be added via DOI, in cases when software has been developed in multiple projects.
  - design documents: the documents such as the Technology Plan, SMP and DMP can be published on Zenodo, and added using their DOIs.

Project Closing Once the project is coming to an end, in addition to the end report data, the following data RSEs are highly encouraged to add:

- testimonials from the project partners or user community
- list of all digital infrastructure used by the project team during the entire project

E Example of the project log

The following example illustrates one possible way of maintaining a project log. The format is intentionally flexible and may be adapted to the needs of the project, provided it offers a clear chronological record of significant project events, decisions, and references to supporting documentation.

Running log for Project XXX

Event	Date	Reference	Notes
Administrative kick-off meeting	2026-03-02	Meeting notes	Project registration completed
Project kick-off meeting	2026-05-09	Slides	Initial workplan agreed
Technology Plan v1	2026-07-18	TechnologyPlan–v1.pdf	Initial version approved
DMP	2026-09-18	Data_Management_Plan.pdf	Initial version approved
Project Review	2026-12-10	Review slides	Continue project
Presentation	2027-01-18	Zenodo DOI	Added to RSD
Software release v1.0	2027-02-05	GitHub release	DOI registered on Zenodo
Research article	2027-03-12	Journal DOI	Added to RSD

2027-03-12 – Research article accepted The publication was accepted, registered in the RSD, and added to the Netherlands eScience Center Zenodo community.

2027-01-24 – SURF infrastructure approved The SURF allocation was approved. The project will use Research Cloud for development and Snellius for production-scale computations.

2027-01-07 – SURF infrastructure request submitted A request for SURF compute and storage resources was submitted based on the agreed Technology Plan.

2026-11-28 – Project meeting In preparation for the project review, the project team met to discuss progress and prepare the presentation slides.

2026-10-25 – Project meeting Meeting with external stakeholders to discuss project progress and next steps.
Agreements:

- Merge the project development branch into the main branch after adding the new tests.
- Prepare a joint presentation for the project review.

2026-06-25 – Project paused The LA was unavailable until 2026-09-01.

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2026-05-09 – Project kick-off meeting
Participants – eScience Center: AB (**SH**), AA (**Lead RSE**), AC (RSE), AD (RSE).
Participants – Project team: FA (LA, TU Delft), PA (PhD candidate, TU Delft), RA (postdoctoral researcher, TU Delft).
Agreements:

- Finalize the Technology Plan before 18 September 2026.
- Submit the SURF infrastructure request before the end of September.
- Schedule fortnightly project meetings with the project team.
- Create the project page in the RSD and populate the initial metadata.

F Internal Procedure of the Project Workshops Approval

The PM sends the LA a workshop proposal template prior to submission. The internal (assessment) procedure consists of several steps, as follows:

1. The LA submits the workshop proposal by email to the project PM. The submitted proposal consists of both scientific part and a financial part.
2. The PM provides the workshop proposal to Finance and DoO; they only assess the financial part. The DoO approves or rejects the financial part of the proposal and immediately informs the PM.
3. The PM only assesses the scientific part of the proposal and gives positive or negative advice to the PD. The PD approves or rejects the scientific part of the proposal and immediately informs the PM.
4. If a part of the proposal is not approved, the PM will contact the LA with a request for adjustment and/or additional information.
5. If both parts of the proposal are approved,
   - the PM archives the approved workshop proposal in the project portfolio folder.
   - The PM informs Finance of the overall positive decision with the request to process it in the financial system.
   - The PM prepares a letter to the LA informing that the workshop proposal has been approved. The letter also contains all further preconditions imposed on the LA and the organization of the workshop.
   - The PM asks the PD to sign the letter, saves the signed copy in the project portfolio folder, informs Finance, and sends a copy to the LA.

Key eScience Roles appearance